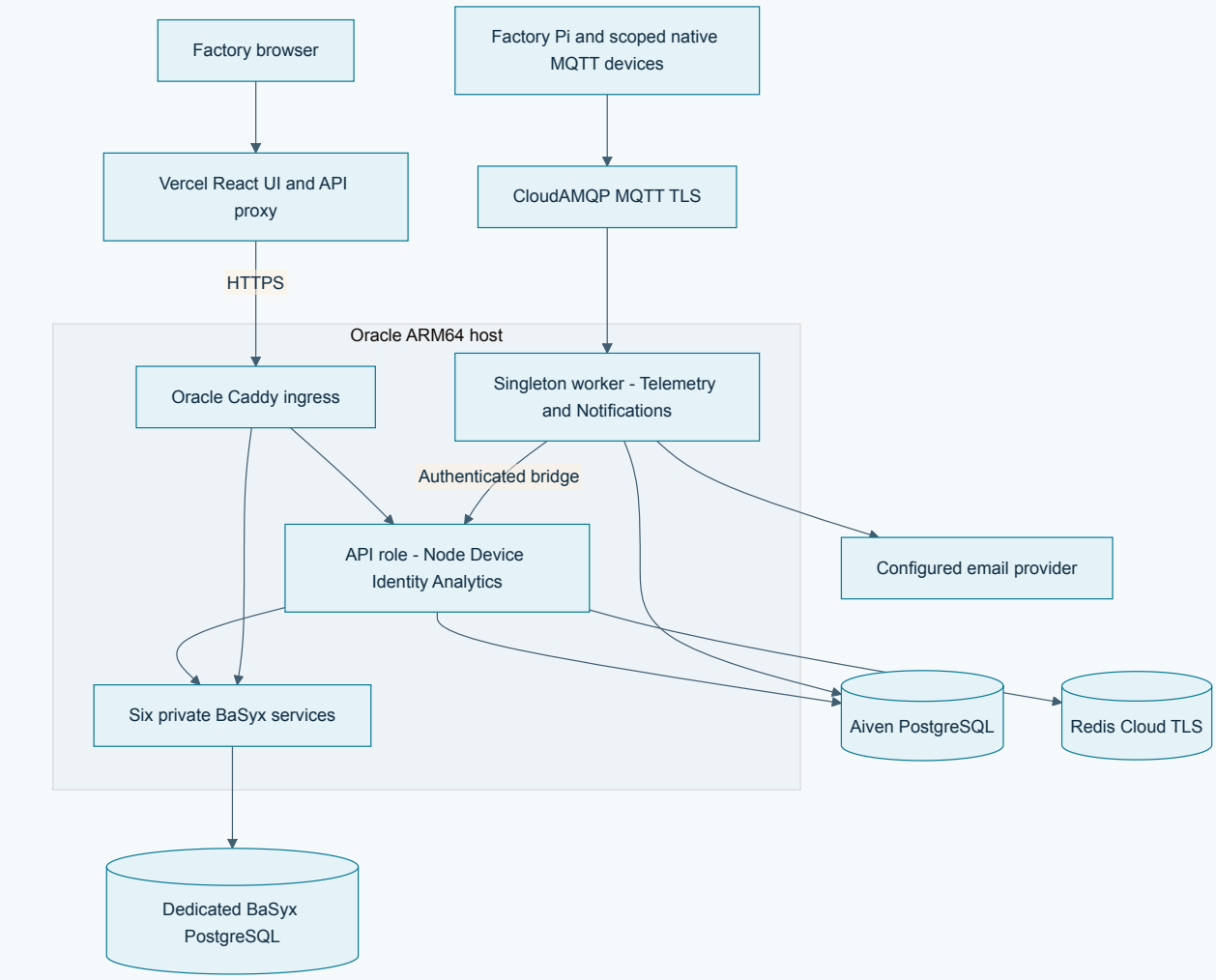
Smart Factory IoT Deployment Guide

The React frontend runs on Vercel. The current backend runs on Oracle as separate API and singleton worker containers from one reviewed ARM64 image, alongside the separately managed BaSyx stack and Caddy ingress. PostgreSQL, Redis, and MQTT retain their managed providers. The previous Render services are suspended.

Deployment map



Deployment map · Open the diagram for a full-size view.

The API role contains Node, DeviceService, IdentityService, and AnalyticsService. The worker role contains TelemetryService and NotificationService. `RENDER_SERVICE_ROLE` remains the role variable for image compatibility; its name does not identify the current hosting provider. Internal service and diagnostic ports remain private or loopback-only.

Local development

Keep the frontend, backend, and edge repositories side by side. Use Docker Compose v2, the repositories' pinned Node/pnpm versions, Python 3.11, and .NET 8 where needed. Configure ignored `.env.local` files from the supplied examples, using synthetic accounts and local dependencies.

```
# Frontend repository: local dashboard, PostgreSQL, Redis, and AAS profile
docker compose --env-file .env.local --profile aas up --build -d --wait

# Backend repository: private .NET services
docker compose --env-file .env.local up --build -d --wait
```

Match the intended service tokens between these local stacks and set `BACKEND_DEPLOYMENT_MODE=compose`. Local physical-device checks require the appropriate broker subscription configuration; a default local RabbitMQ consumer does not receive a physical Pi's CloudAMQP messages automatically.

Runtime configuration boundaries

Boundary	Configuration and constraints
Browser build	Public frontend routing only; no database, MQTT, AAS, or AI service secrets
API role	Dashboard database, Redis, session signing, service tokens, AAS OAuth, exact allowed origins
Worker role	Telemetry database, stable MQTT client ID, scoped broker credentials, service tokens, API origin, mail settings
BaSyx stack	Independent component URLs, OIDC client credentials, dedicated database, and private component ports
Edge	Local allowlists, commissioned protocol mappings, scoped MQTT identities, and durable spool

Use verified TLS for PostgreSQL, Redis, and cloud MQTT. Keep `.env.api` and `.env.worker` protected and untracked; the Oracle Compose raw env-file mode requires Compose 2.30 or newer. Preserve literal credentials during transfer without printing resolved configuration.

Image update sequence

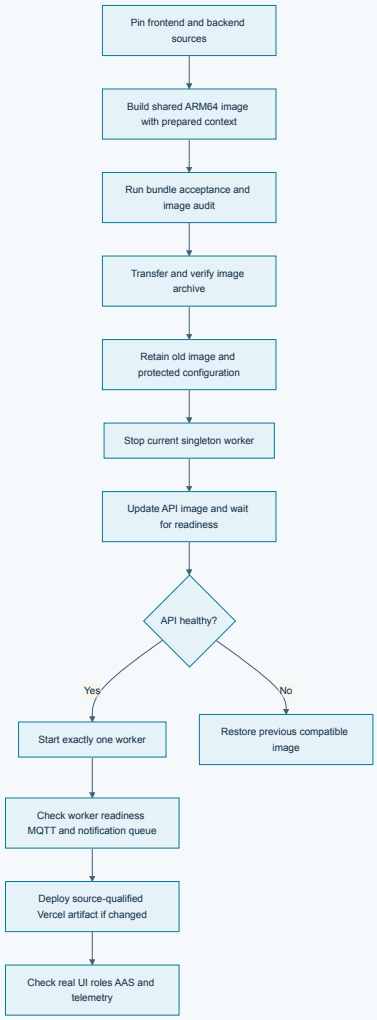


Image update sequence · Open the diagram for a full-size view.

Build from the frontend checkout with the prepared backend context and `linux/arm64`, then run the repository's bundle smoke and container-audit scripts. Transfer the tested image and preserve the installed configuration. Backend image updates use the manual Oracle procedure; the current Vercel workflow deploys a source revision only after its matching main CI succeeds.

Do not run multiple production workers with the same stable MQTT consumer identity. During any provider rollback, stop the active worker before starting its replacement. Routine updates do not replace the private environment exports or create new databases.

Health and acceptance

Check	Purpose
API `/health/live`	API process liveness
API `/health/ready`	Dashboard database, Redis, and required private services
Worker readiness and Supervisor state	Telemetry readiness plus both worker processes
Real account through Vercel	Secure session cookie, role delegation, logout, allowed and denied actions
AAS component descriptions	Authenticated access to all six BaSyx components
Commissioned telemetry path	Fresh device values, correct gateway attribution, stable session, draining queues
Notifications	Durable inbox and configured provider state; acceptance is not confirmed mailbox delivery

Use simulated equipment for ordinary acceptance. Hardware commissioning and physical-motion verification are separate site activities. Configuration acknowledgements, broker publication, and controller serial acknowledgements are different events and must not be treated as proof of machine motion.

Rollback and recovery

Retain the last accepted image, ingress configuration, and protected runtime settings. Restore a compatible backend image, verify API readiness, and start one worker. Restore the previous Vercel artifact if its proxy or contract changed. Schema rollback or database restoration requires its own reviewed plan.

Managed PostgreSQL holds application state; Redis Pub/Sub is transient transport. Edge spools and the telemetry outbox support retry after interruption. Monitor backlog, spool capacity, database pools, and failed notification jobs. A shared Oracle host is one failure domain and does not establish high availability.

Source runbooks

- Current Oracle API and singleton-worker runbook
- BaSyx runtime
- Architecture, API flows, and system diagrams

Source review: 21 October 2026. The portfolio guide documents the release process and does not perform a deployment.